

# Medical Vision-Language Models:

From pre-training to finetuning and adaptation

MICCAI 2026 Tutorial on Multimodal Foundation Models for Biomedical Imaging  
(MULTIFOUND)

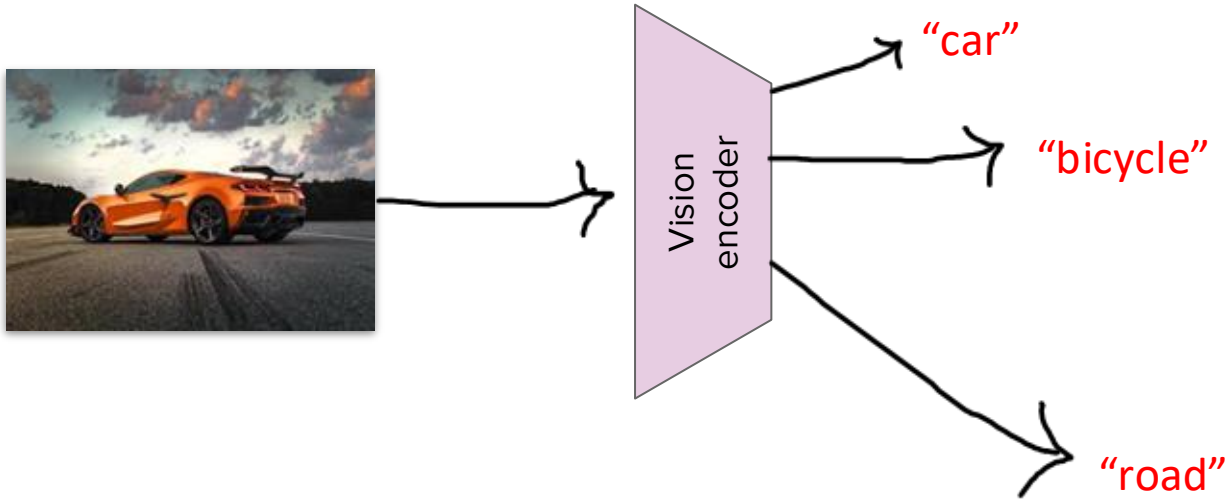
Parts 2

Ismail Ben Ayed

# Part 2

## Contrastive Vision-Language Models for Medical Imaging: From Generalist to Specialist

# From unimodal to multimodal: Standard concept classifiers in vision



# From unimodal to multimodal:

Also the standard way in medical image analysis

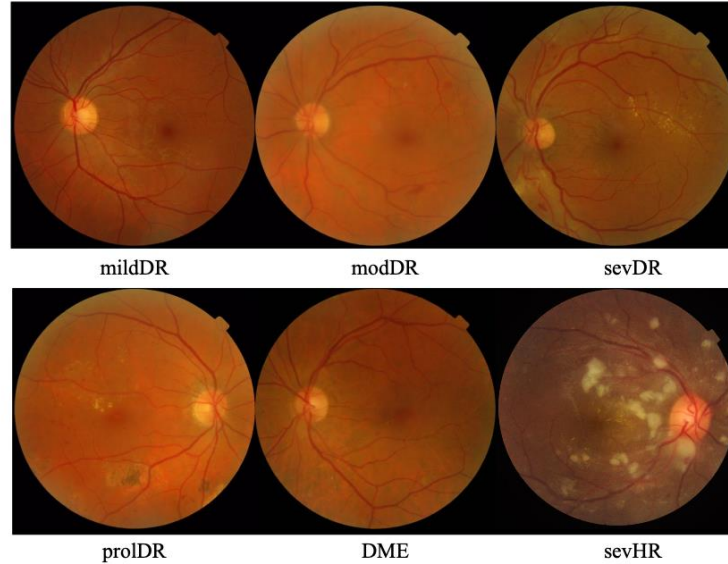
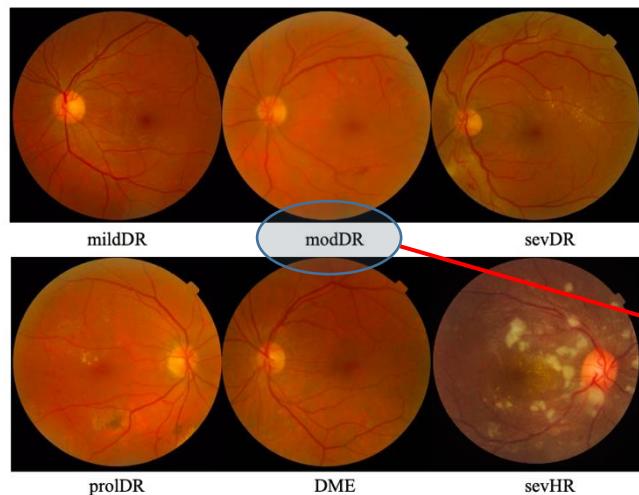


Figure from Silva-Rodriguez et *al.* A Foundation Language-Image model of the Retina (FLAIR): Encoding expert knowledge in text supervision, ***Medical Image Analysis***, 2025

# The model has no clue about the underlying biomarkers (hence poor interpretability)



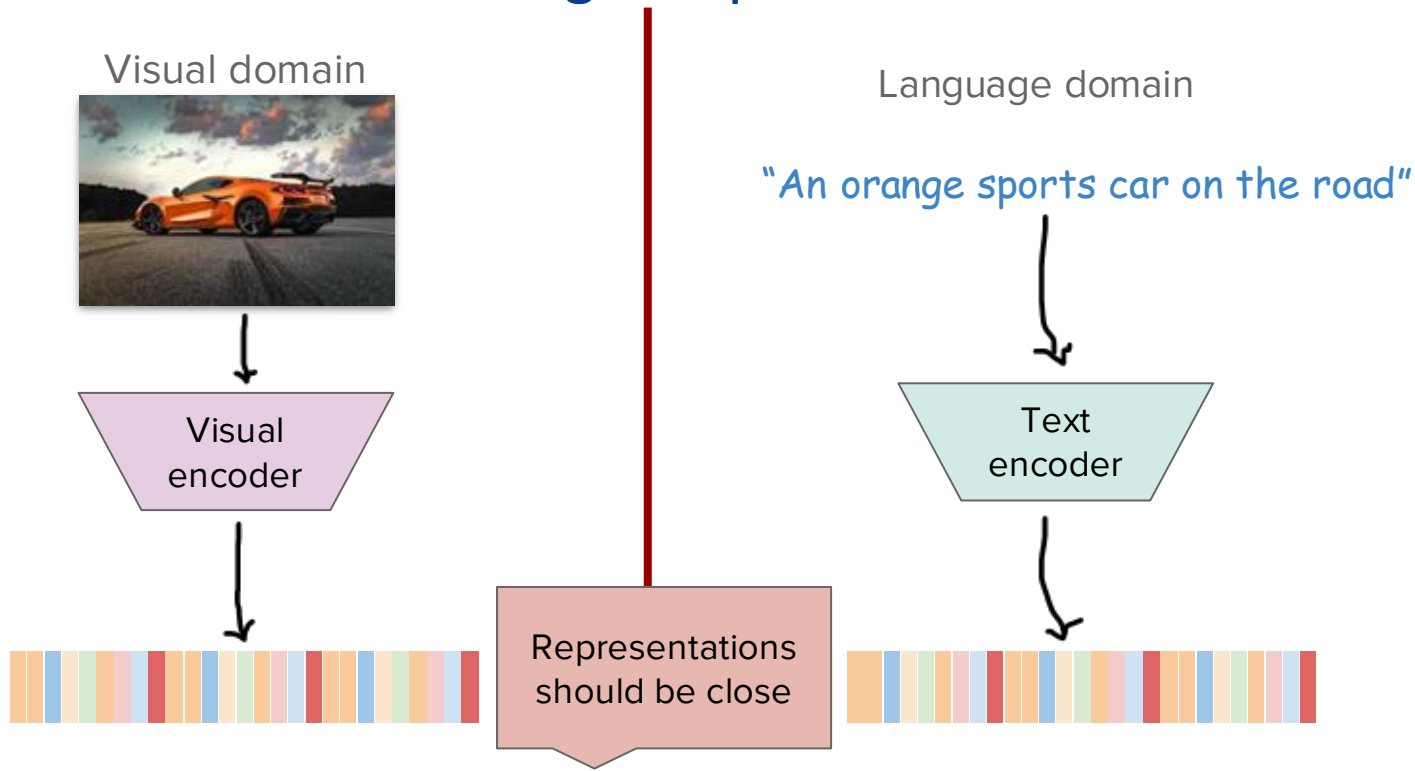
This is what human experts know and  
how they make their decisions:

“retinal haemorrhages in a few quadrants”

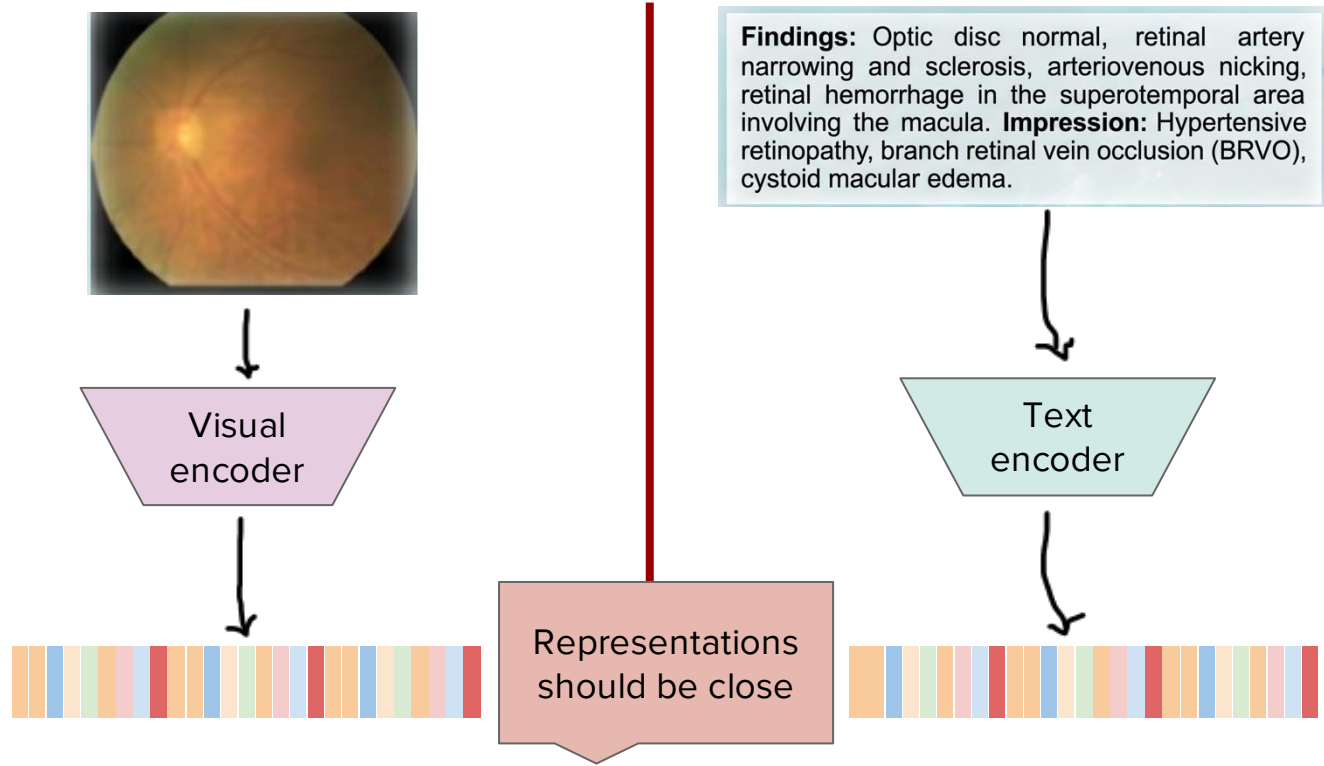
“many haemorrhages”

“Cotton wool spots”

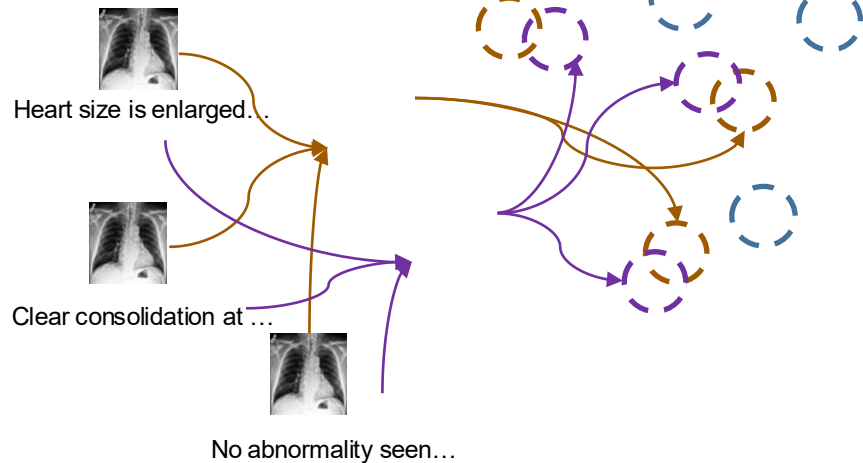
# Foundational Vision-Language Models (VLMs) are transforming computer vision



## And medical imaging...



# Medical vision-language models



**CONVIRT**: Zhang et al., Medical Visual Representations from Paired Images and Text, MLHC 2022

**MedCLIP**: Wang et al., Contrastive Learning from Unpaired medical images and text, EMNLP 2022

**Quilt-1M**: Ikezogwo et al., One Million Image-Text Pairs for Histopathology, NeurIPS 2023

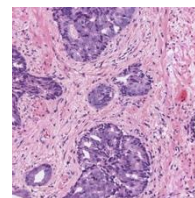
**FLAIR**: Silva-Rodríguez et al., A Foundation Language-Image Model of the Retina, MedIA 2025

...

MedCLIP



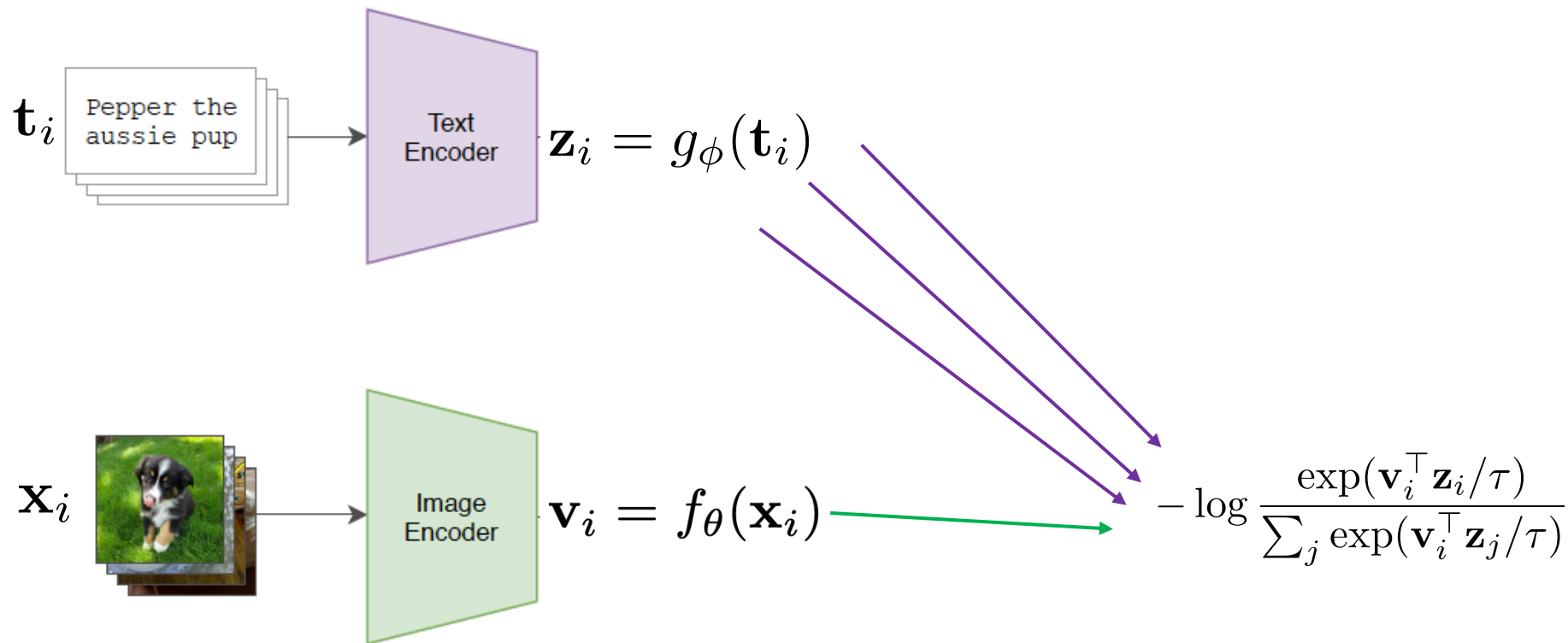
Quilt-1M



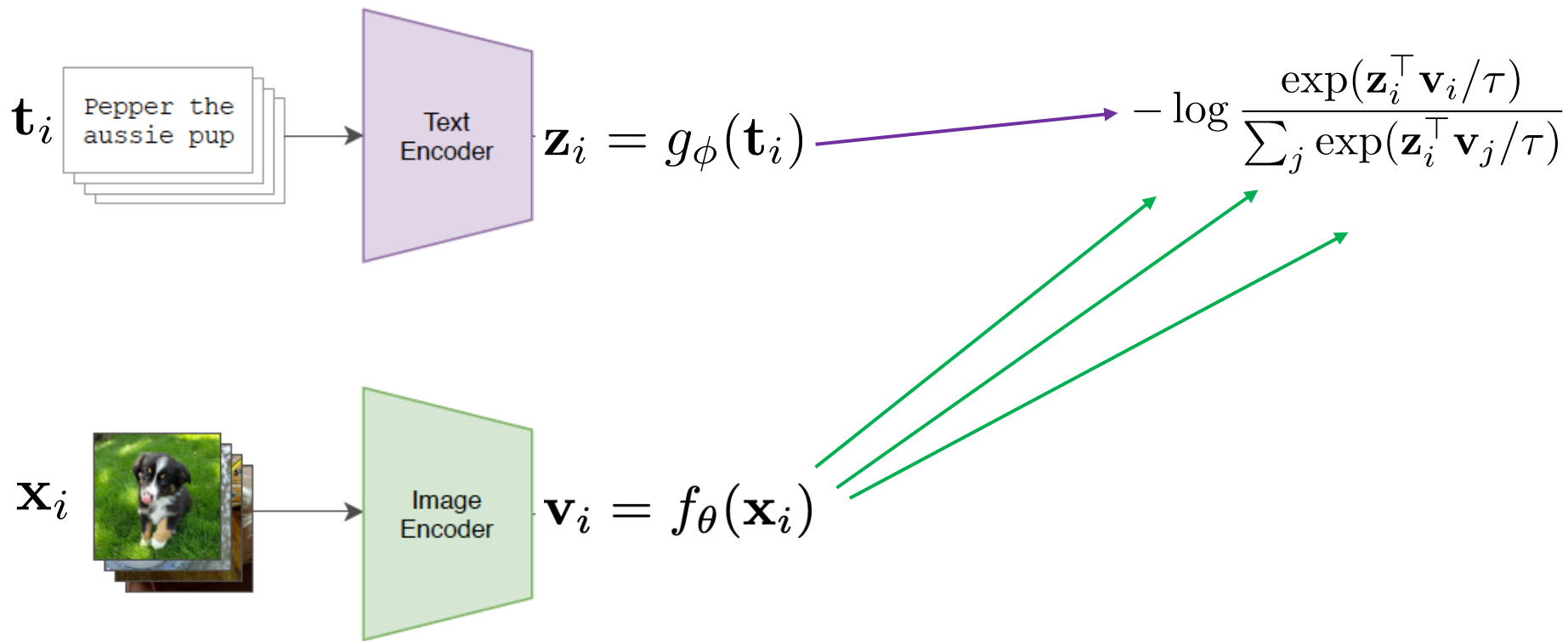
FLAIR



# Contrastive Language-Image Pre-training (CLIP)

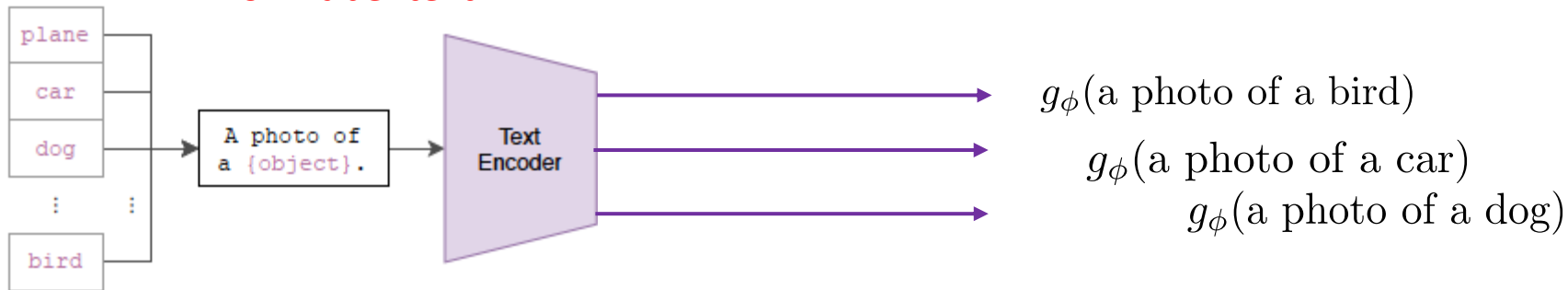


# Contrastive Language-Image Pre-training (CLIP)



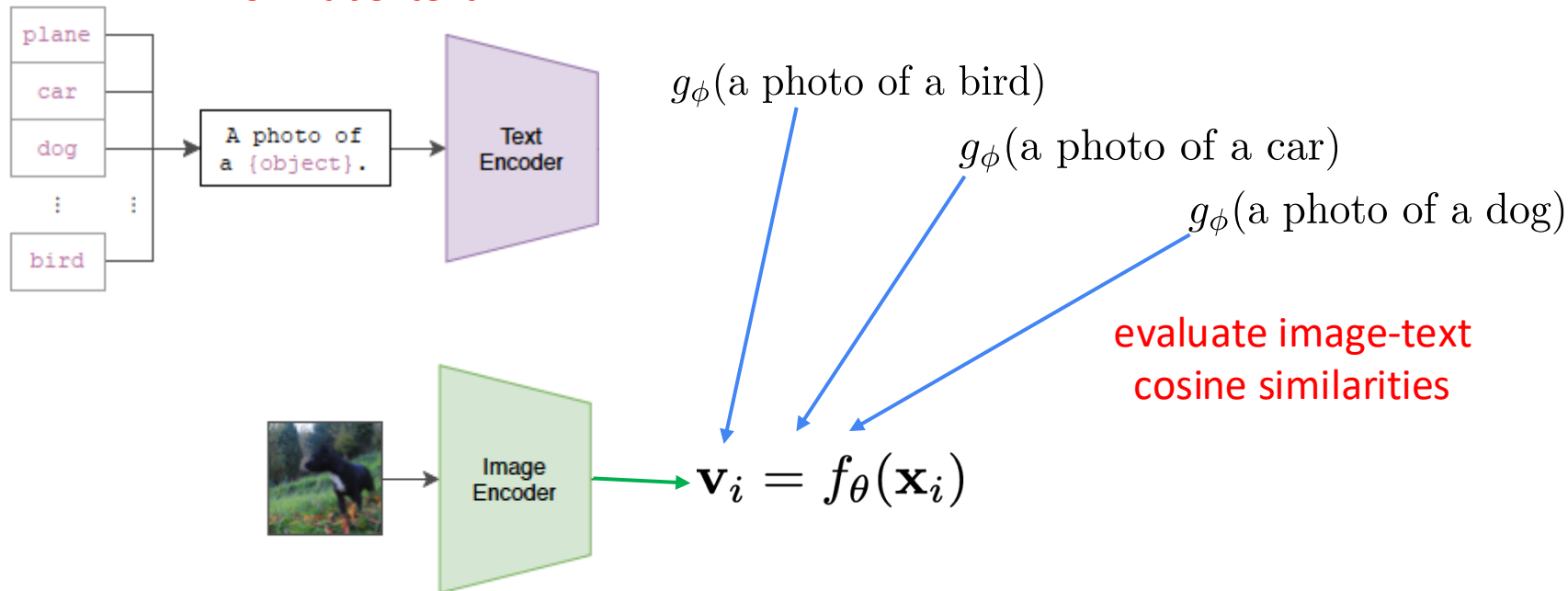
# “Zero-shot” prediction for new tasks

create “class prototypes”  
From label text



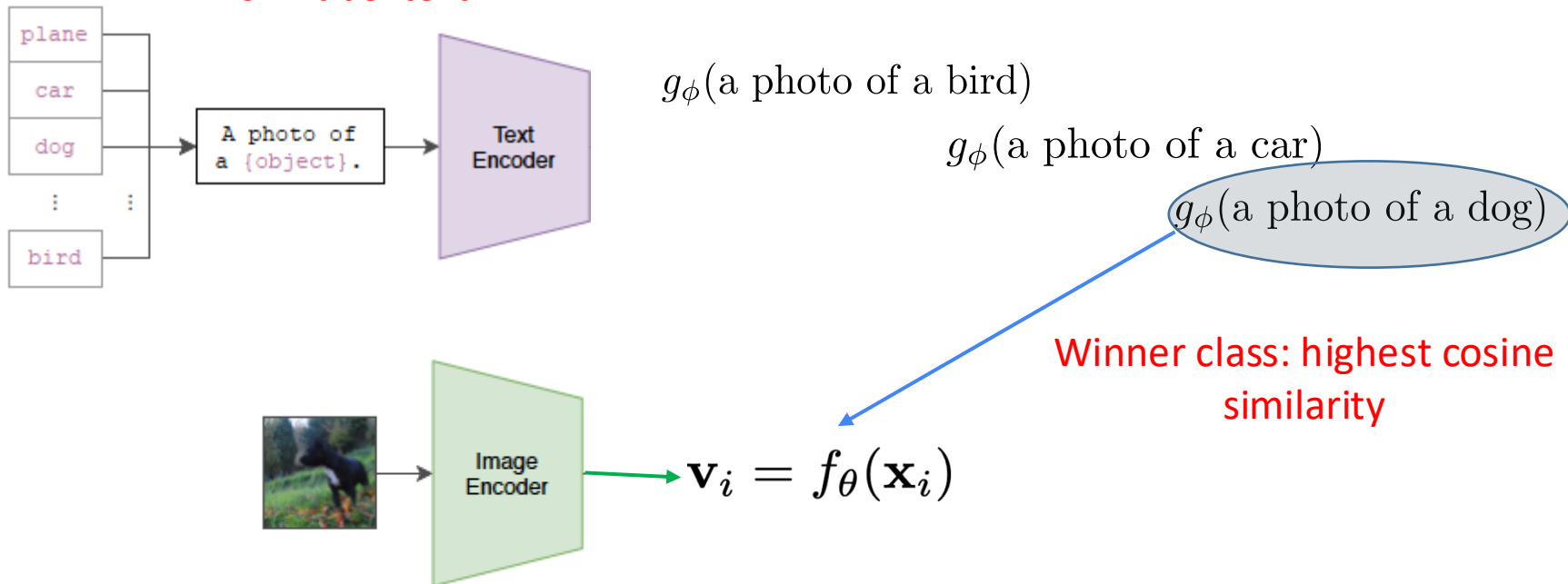
# “Zero-shot” prediction for new tasks

create “class prototypes”  
From label text



# “Zero-shot” prediction for new tasks

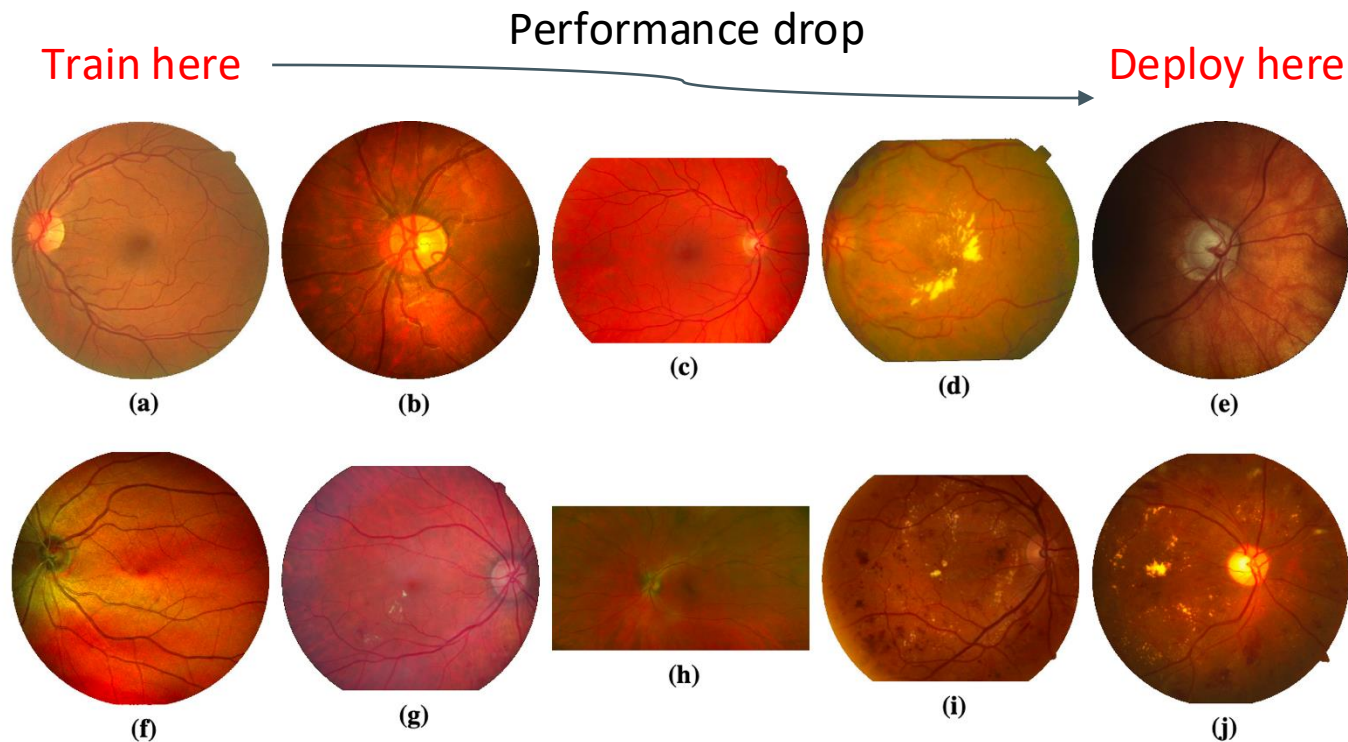
create “class prototypes”  
From label text



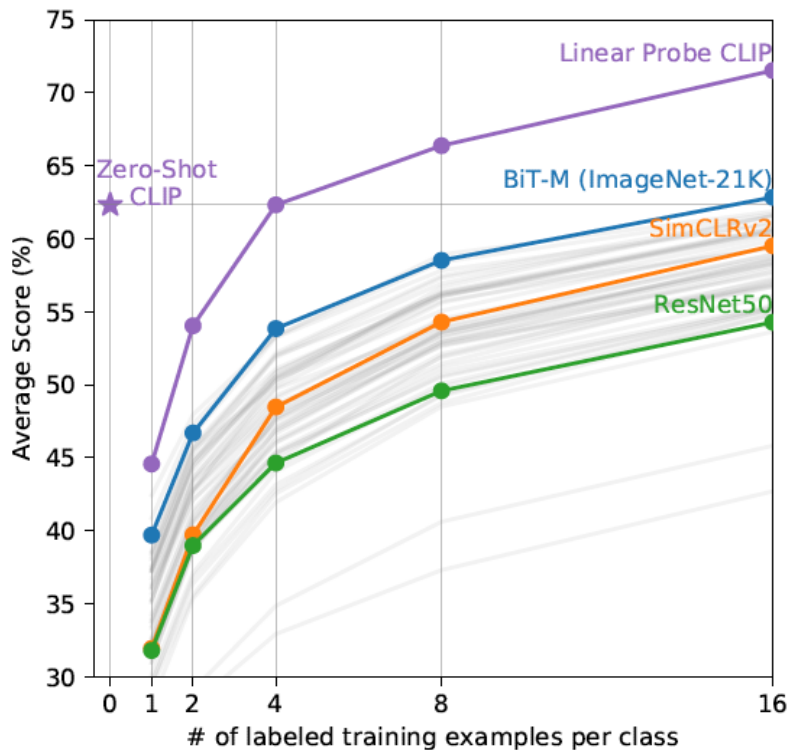
# Promising zero-shot generalization (in comparison to standard task-specific learning)

|                 | Dataset Examples                                                                   | ImageNet<br>ResNet101 | Zero-Shot<br>CLIP | $\Delta$ Score |
|-----------------|------------------------------------------------------------------------------------|-----------------------|-------------------|----------------|
| ImageNet        |  | 76.2                  | 76.2              | 0%             |
| ImageNetV2      |  | 64.3                  | 70.1              | +5.8%          |
| ImageNet-R      |  | 37.7                  | 88.9              | +51.2%         |
| ObjectNet       |  | 32.6                  | 72.3              | +39.7%         |
| ImageNet Sketch |  | 25.2                  | 60.2              | +35.0%         |
| ImageNet-A      |  | 2.7                   | 77.1              | +74.4%         |

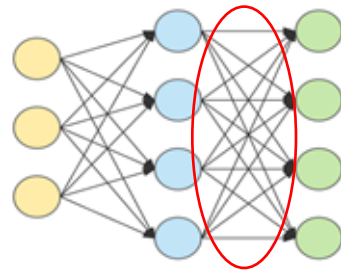
# Performance drop due to data variations is a serious issue in medical imaging practice as well



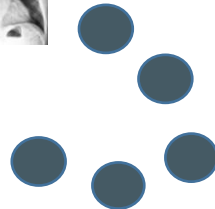
# Beyond zero-shot: Fast adaptation on your own data and task



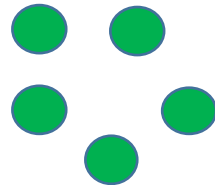
Few-shot fine-tuning (linear probe)



Class 1: pneumonia



Class 2: normal



# Beyond robustness: Such knowledge enables interpretability!

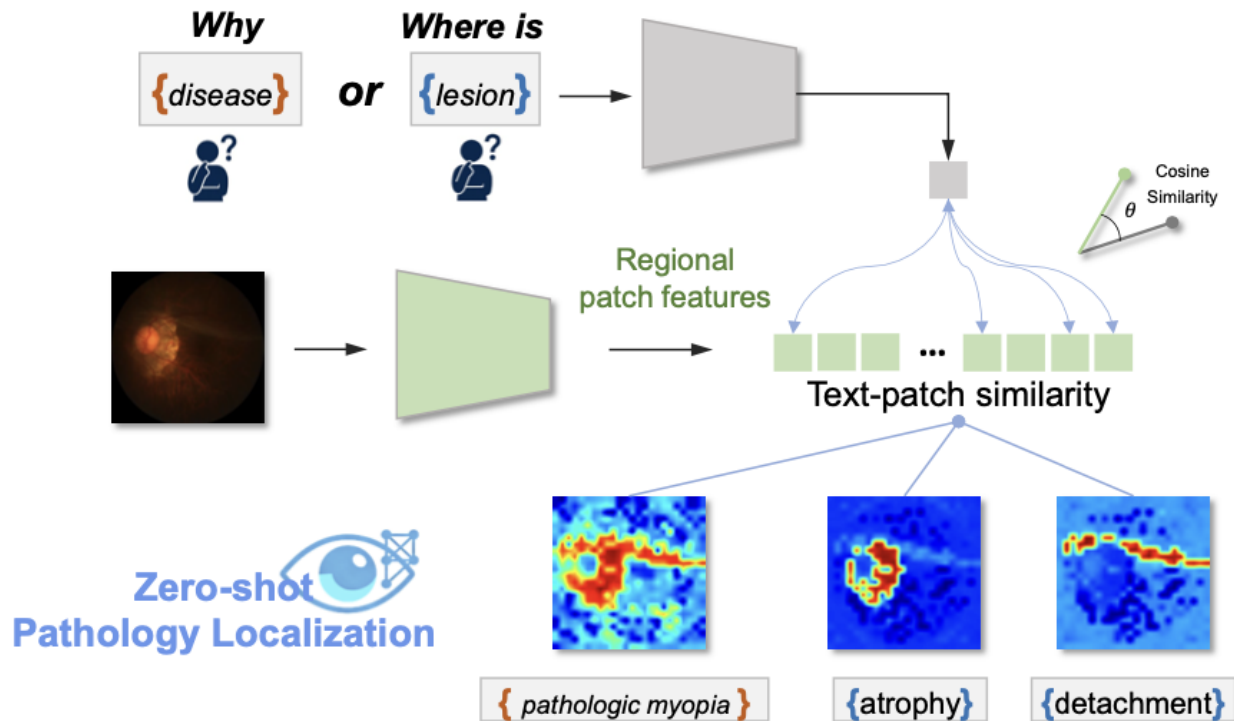


Figure from Guo et al., Native Intelligence Emerges from Large-Scale Clinical Practice: A Retinal Foundation Model with Deployment Efficiency, [arXiv, Dec. 2025](#)

# Beyond robustness: Such knowledge enables interpretability!

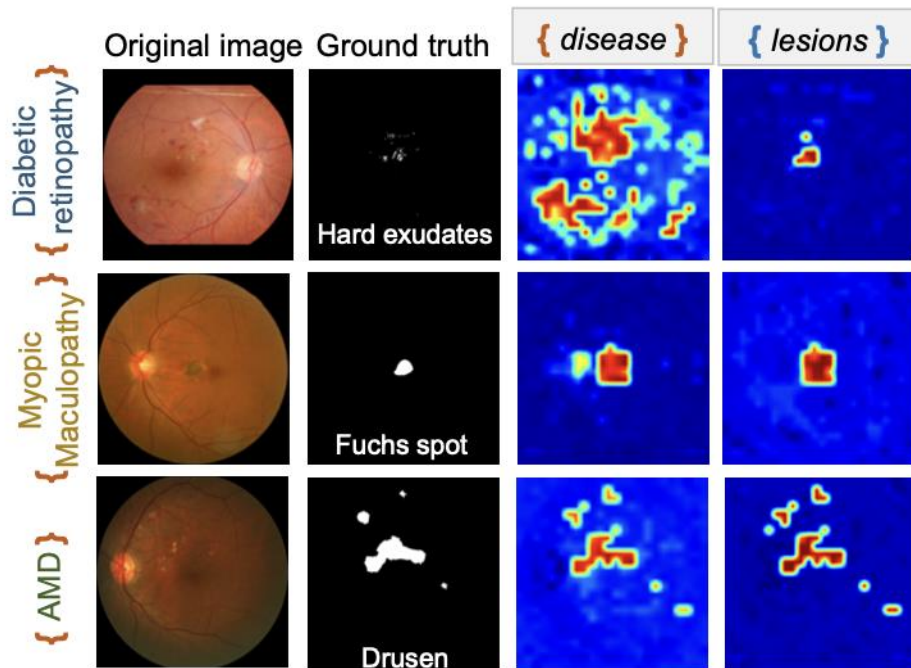


Fig. from Guo et al., Native Intelligence Emerges from Large-Scale Clinical Practice: A Retinal Foundation Model with Deployment Efficiency, **arXiv, Dec. 2025**

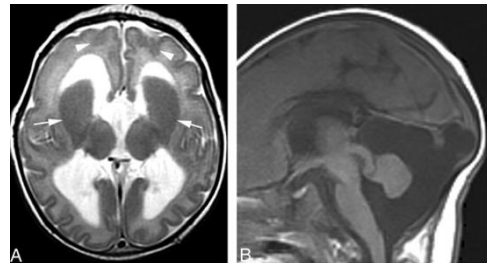
# Text-image data is available in some domains (e.g. radiology)



Severe **cardiomegaly** is noted in the image with enlarged...



Radiograph shows **pleural effusion** in the right lobe...



**CLINICAL HISTORY:** 6 wk old, term, developmental delay, microcephaly, seizures

**FINDINGS:**

**BRAIN PARENCHYMA:** No acute infarct or hemorrhage. Markedly enlarged caudate with absent internal capsule, resulting in fused striata bilaterally.

**CORTEX:** Bifrontal band heterotopias are seen as well as bioccipital pachygyria.

**SULCATION:** Delayed for age.

**WHITE MATTER:** Reduced volume. Myelination is delayed for age.

**CORPUS CALLOSUM:** Thinned.

**VENTRICLES:** Enlarged.

**CEREBELLUM:** Hypoplastic.

**VERMIS:** Hypoplastic.

**BRAINSTEM:** Hypoplastic pons.

**MRS:**

NAA:choline ratio: Normal.

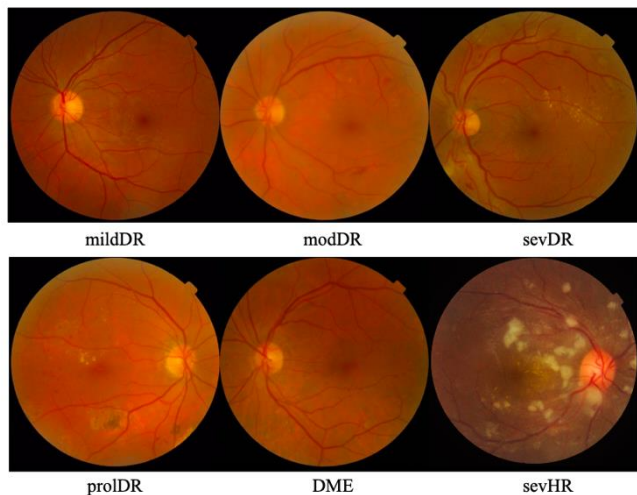
Lactate Peak: Absent.

**IMPRESSION:**

Combination of abnormal basal ganglia, malformations of cortical development, microcephaly, and a small cerebellum/brainstem are compatible with a tubulinopathy.

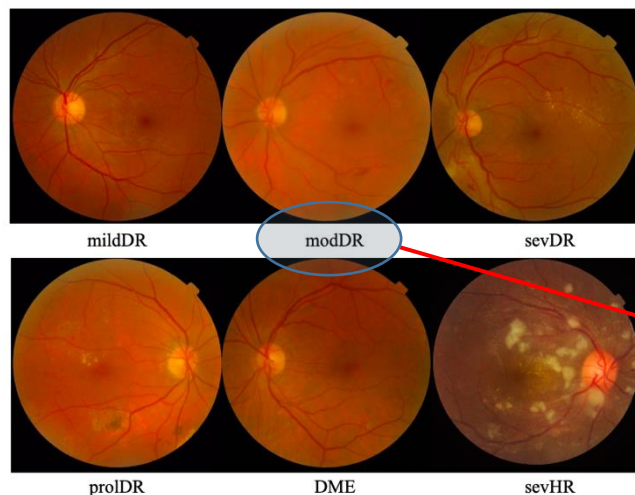
C

# Not always: In ophthalmology most of available public data is categorical data



Silva-Rodriguez et al. A Foundation LAnguage-Image model of the Retina (FLAIR): Encoding expert knowledge in text supervision, *Medical Image Analysis*, 2025

# Can we leverage knowledge from separate text data (e.g., the clinical literature)?



American Academy of Ophthalmology  
Wilkinson et al., 2003

“retinal haemorrhages in a few quadrants”

“many haemorrhages”

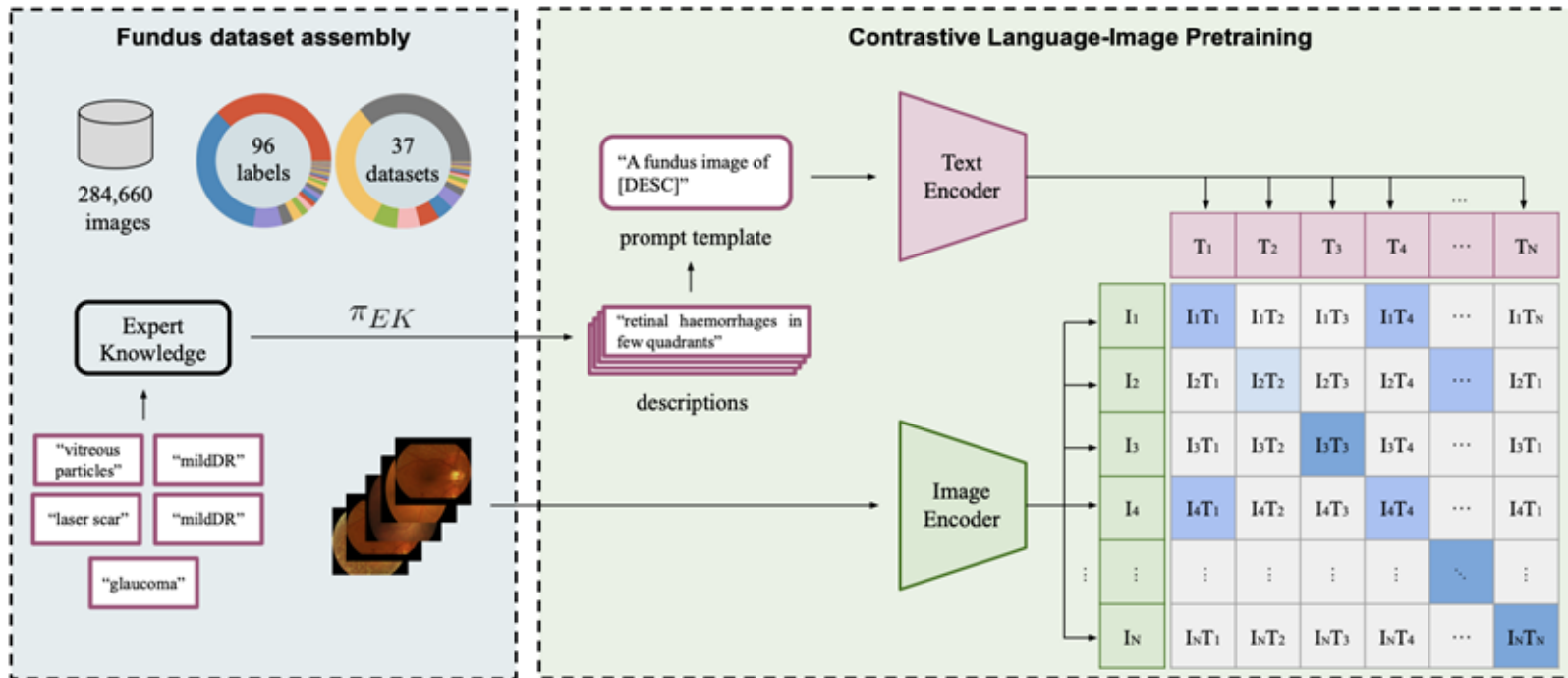
“Cotton wool spots”

# A Foundation LAnguage-Image model of the Retina (FLAIR)

| Category                                          | Domain Knowledge descriptor                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no diabetic retinopathy                           | "no relevant haemorrhages, microaneurysms or exudates" / "no microaneurysms" / "no referable lesions"                                                                                              |
| mild diabetic retinopathy                         | "few microaneurysms" / "few hard exudates" / "few retinal haemorrhages"                                                                                                                            |
| moderate diabetic retinopathy                     | "retinal haemorrhages in few quadrants" / "many haemorrhages" / "cotton wool spots"                                                                                                                |
| severe diabetic retinopathy                       | "severe haemorrhages in all four quadrants" / "venous beading" / "intraretinal microvascular abnormalities"                                                                                        |
| proliferative diabetic retinopathy                | "diabetic retinopathy with neovascularization at the disk" / "neovascularization"                                                                                                                  |
| diabetic macular edema                            | "macular edema" / "presence of exudates" / "leakage of fluid within the central macula from microaneurysms" / "presence of exudates within the radius of one disc diameter from the macula center" |
| no referable diabetic macular edema               | "no apparent exudates"                                                                                                                                                                             |
| hard exudates                                     | "small white or yellowish deposits with sharp margins" / "bright lesion"                                                                                                                           |
| soft exudates                                     | "pale yellow or white areas with ill-defined edges" / "cotton-wool spot" / "small, whitish or grey, cloud-like, linear or serpentine, slightly elevated lesions with fimbriated edges"             |
| microaneurysms                                    | "small red dots"                                                                                                                                                                                   |
| haemorrhages                                      | "dense, dark red, sharply outlined lesion"                                                                                                                                                         |
| non clinically significant diabetic macular edema | "presence of exudates outside the radius of one disc diameter from the macula center" / "presence of exudates"                                                                                     |
| age-related macular degeneration                  | "many small drusen" / "few medium-sized drusen" / "large drusen"                                                                                                                                   |

Silva-Rodriguez et al. A Foundation LAnguage-Image model of the Retina (FLAIR): Encoding expert knowledge in text supervision, *Medical Image Analysis*, 2025

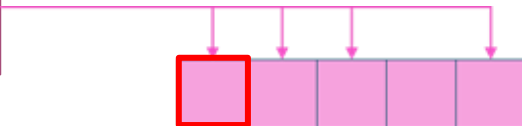
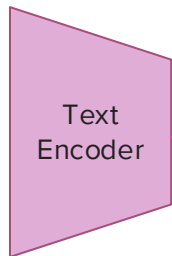
# A Foundation LAnguage-Image model of the Retina (FLAIR)



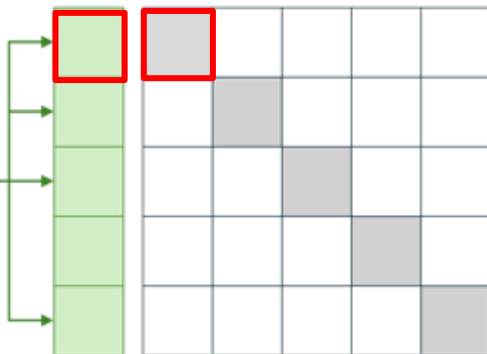
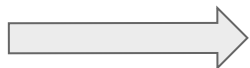
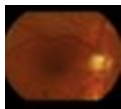
Silva-Rodriguez et al. A Foundation LAnguage-Image model of the Retina (FLAIR): Encoding expert knowledge in text supervision, *Medical Image Analysis*, 2025

# A Foundation LAnguage-Image model of the Retina (FLAIR)

"A fundus image of possible signs of hemorrhage with blot, dot, or flame-shaped"

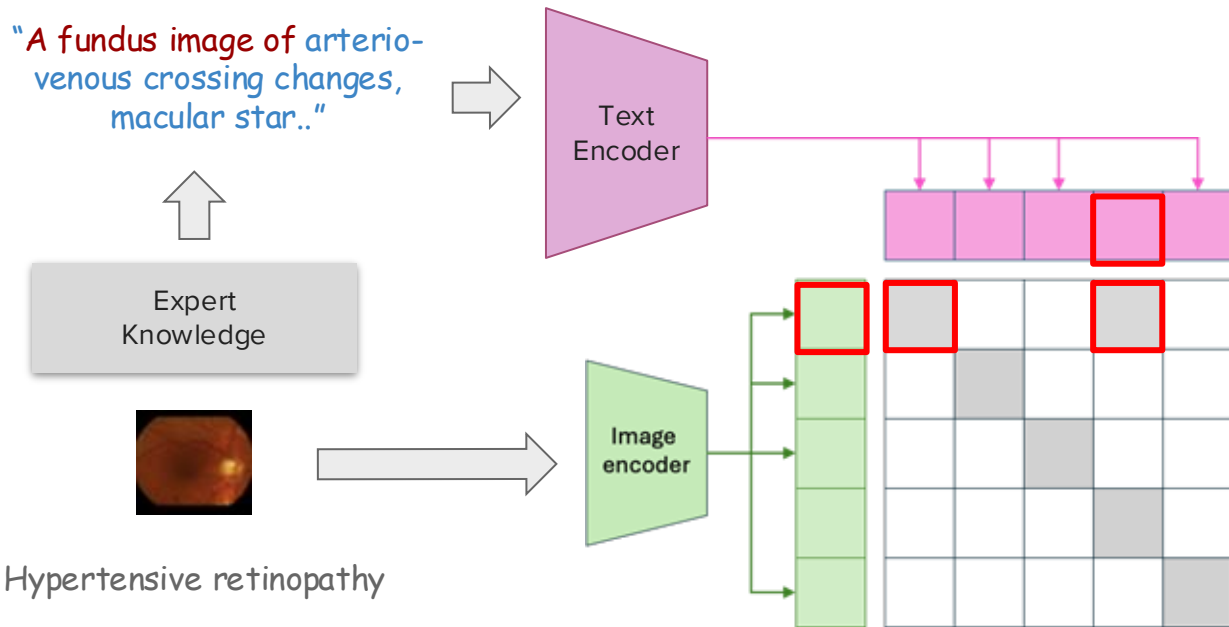


This would be standard CLIP



Hypertensive retinopathy

# A Foundation LAnguage-Image model of the Retina (FLAIR)



What we have now:

$$\mathcal{L}^v = - \sum_{i \in \mathcal{B}} \frac{1}{|P_{\mathcal{T}_B}(i)|} \sum_{i' \in P_{\mathcal{T}_B}(i)} \log \frac{\exp(\mathbf{v}_i^\top \mathbf{z}_{i'} / \tau)}{\sum_{j \in \mathcal{T}_B} \exp(\mathbf{v}_i^\top \mathbf{z}_j / \tau)}$$

# A Foundation LAnguage-Image model of the Retina (FLAIR)

We go through multiple text prompts  
for same class

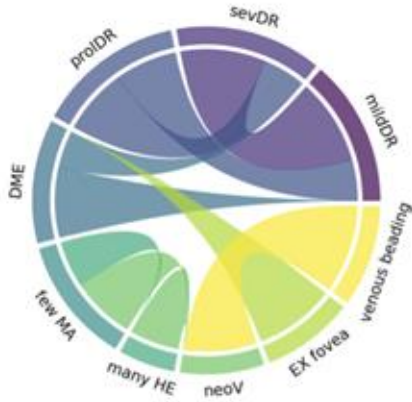
E.g. **‘Hypertensive retinopathy’**

- "possible signs of hemorrhage with blot, dot, or flame-shaped"
- "possible presence of microaneurysm, cotton-wool spot, or hard exudate"
- "arteriolar narrowing"
- "vascular wall changes"
- "optic disk edema"

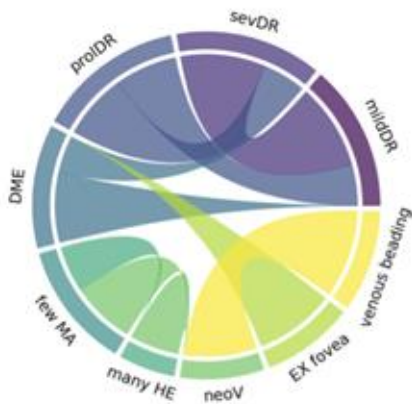
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# A Foundation LAnguage-Image model of the Retina (FLAIR)



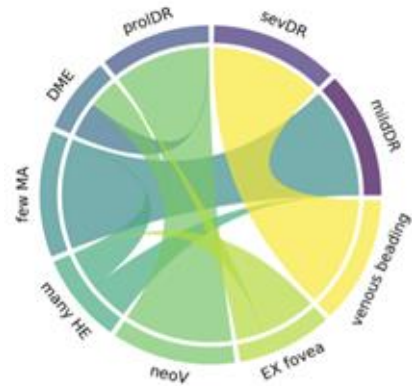
# A Foundation LAnguage-Image model of the Retina (FLAIR)



CLIP focuses on general medical relations.



Adding domain specific-knowledge



E.g., “**neovascularization**” is the differential sign for **proDR**.

# Some interesting results on retinal imaging tasks

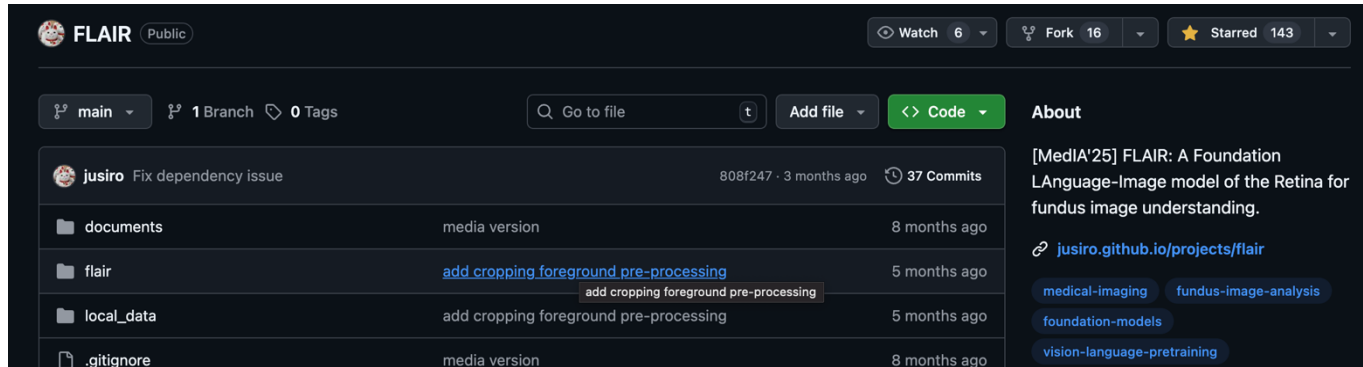
Zhou et al., A foundation model for generalizable disease detection from retinal images, *Nature* 2023 (self-supervised learning)

| (a) <i>Zero-shot</i>      |          | MESSIDOR     | FIVES        | REFUGE       | 20x3         | ODIR <sub>200x3</sub> | MMAC         | Avg.         |
|---------------------------|----------|--------------|--------------|--------------|--------------|-----------------------|--------------|--------------|
| CLIP                      | ViT-B/32 | 0.200        | 0.256        | 0.433        | 0.333        | 0.480                 | 0.183        | 0.314        |
| BiomedCLIP                | ViT-B/16 | 0.207        | 0.415        | 0.624        | 0.617        | 0.583                 | 0.274        | 0.453        |
| FLAIR                     | RN50     | <b>0.604</b> | <b>0.735</b> | <b>0.883</b> | <b>0.983</b> | <b>0.667</b>          | <b>0.400</b> | <b>0.712</b> |
| (b) <i>Linear Probing</i> |          |              |              |              |              |                       |              |              |
| ImageNet                  | RN50     | 0.424        | 0.741        | 0.733        | 0.983        | 0.887                 | 0.631        | 0.733        |
| CLIP                      | ViT-B/32 | 0.491        | 0.800        | 0.720        | 0.950        | 0.917                 | 0.642        | 0.753        |
| BiomedCLIP                | ViT-B/16 | 0.433        | 0.654        | 0.776        | 0.866        | 0.883                 | 0.678        | 0.715        |
| RETFound                  | ViT-B/16 | 0.457        | 0.765        | 0.747        | 0.950        | 0.887                 | 0.547        | 0.725        |
| FLAIR                     | RN50     | <b>0.719</b> | <b>0.879</b> | <b>0.843</b> | <b>1.000</b> | <b>0.935</b>          | <b>0.740</b> | <b>0.852</b> |

!!!

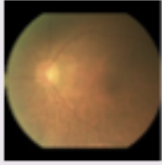
# FLAIR: Code and data publicly available

<https://github.com/jusiro/FLAIR>



Silva-Rodriguez et al. A Foundation Language-Image model of the Retina (FLAIR): Encoding expert knowledge in text supervision, *Medical Image Analysis*, 2025

# Recent image-text data for the retina



**Findings:** Optic disc normal, retinal artery narrowing and sclerosis, arteriovenous nicking, retinal hemorrhage in the superotemporal area involving the macula. **Impression:** Hypertensive retinopathy, branch retinal vein occlusion (BRVO), cystoid macular edema.



**Findings:** Vitreous hemorrhage, extensive retinal hemorrhage, hard exudates. **Impression:** Proliferative diabetic retinopathy (PDR). **Medical History:** Diabetes for over 10 years.



**TM500k**

162 centers  
251,243 patients  
485,980 images



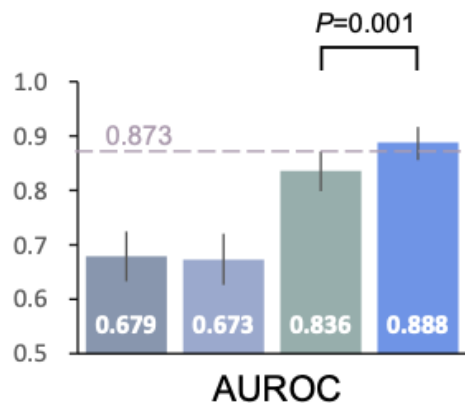
**Findings:** The optic disc is pale red with clear borders, cup-to-disc ratio is enlarged to 0.5, no abnormalities observed in the macular region. **Impression:** Suspected glaucoma.

Fig. from Guo et al., Native Intelligence Emerges from Large-Scale Clinical Practice: A Retinal Foundation Model with Deployment Efficiency, **arXiv, Dec. 2025**

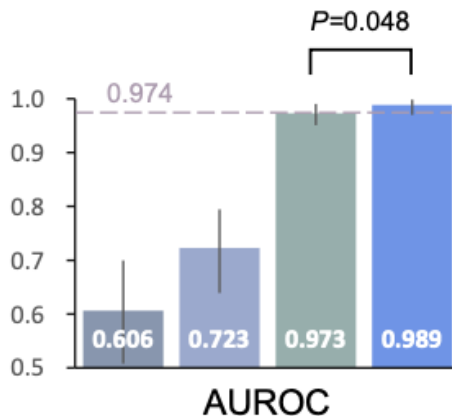
# Some results confirm the importance of domain knowledge

OpenAI-CLIP BiomedCLIP FLAIR ReVision

Diabetic retinopathy



Glaucoma



AMD

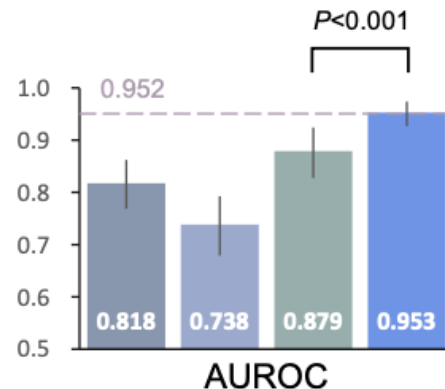


Fig. from Guo et al., Native Intelligence Emerges from Large-Scale Clinical Practice: A Retinal Foundation Model with Deployment Efficiency, [arXiv](#), Dec. 2025

# Impressive interpretability emerges from scale

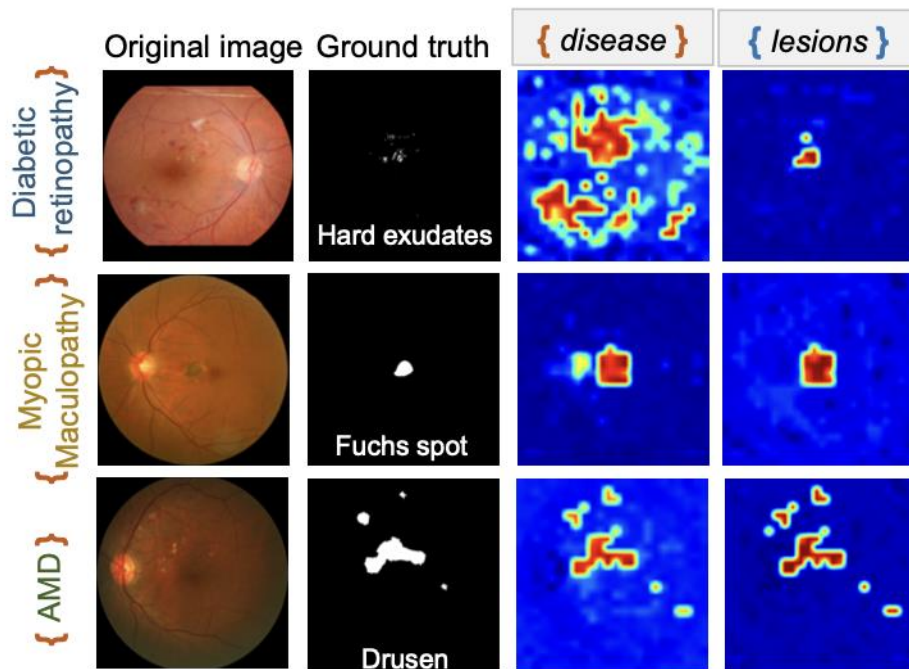
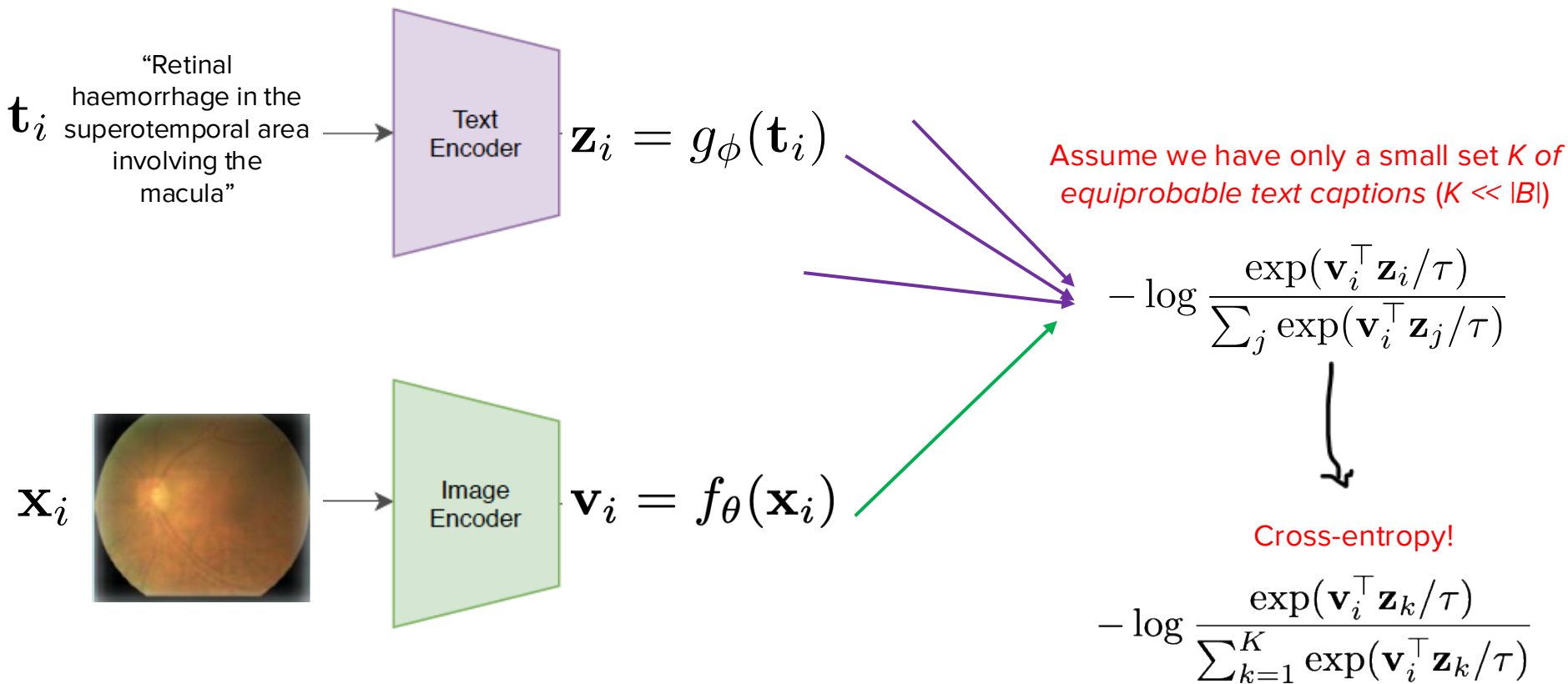


Fig. from Guo et al., Native Intelligence Emerges from Large-Scale Clinical Practice: A Retinal Foundation Model with Deployment Efficiency, **arXiv, Dec. 2025**

**Beyond contrasting all (CLIP, FLAIR):**  
Multi-labels and medical concept hierarchies/relationships

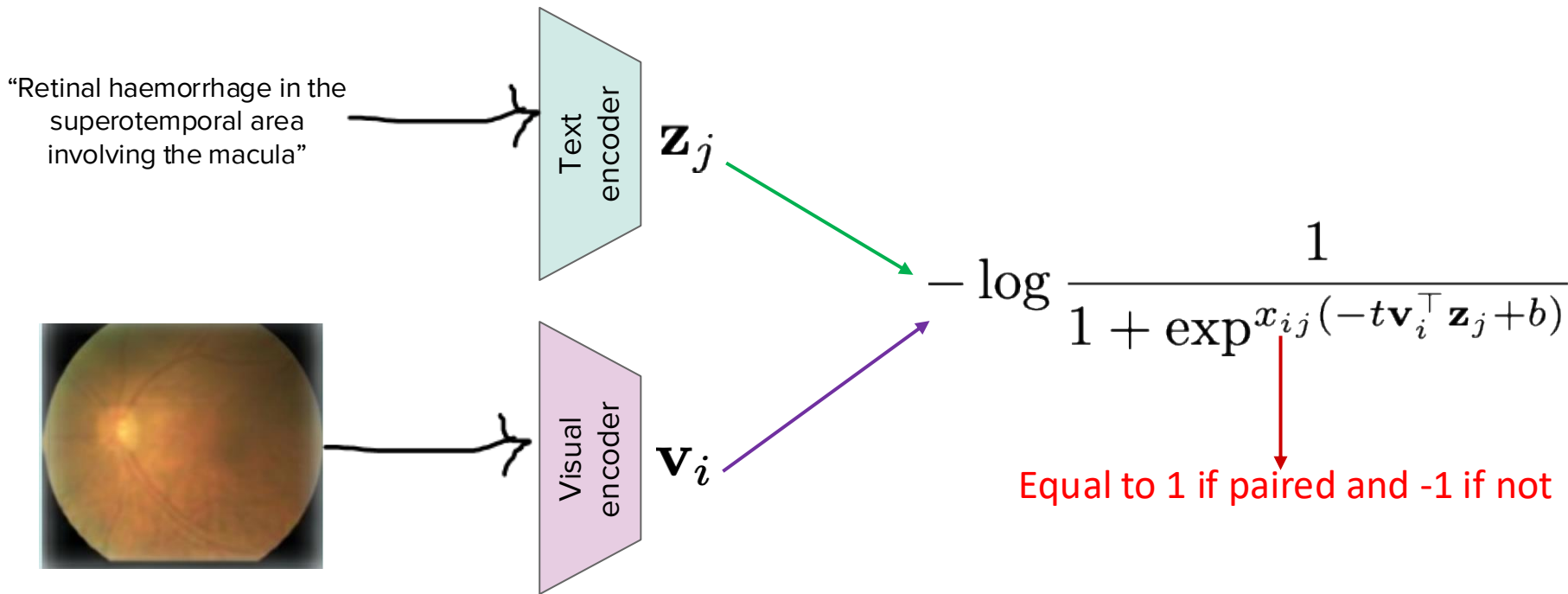
# CLIP is a multi-logit cross-entropy (CE) loss (Almost)



# CLIP/FLAIR might make strongly related concepts mutually exclusive

| Category                                          | Domain Knowledge descriptor                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no diabetic retinopathy                           | "no relevant haemorrhages, microaneurysms or exudates" / "no microaneurysms" / "no referable lesions"                                                                                              |
| mild diabetic retinopathy                         | "few microaneurysms" / "few hard exudates" / "few retinal haemorrhages"                                                                                                                            |
| moderate diabetic retinopathy                     | "retinal haemorrhages in few quadrants" / "many haemorrhages" / "cotton wool spots"                                                                                                                |
| severe diabetic retinopathy                       | "severe haemorrhages in all four quadrants" / "venous beading" / "intraretinal microvascular abnormalities"                                                                                        |
| proliferative diabetic retinopathy                | "diabetic retinopathy with neovascularization at the disk" / "neovascularization"                                                                                                                  |
| diabetic macular edema                            | "macular edema" / "presence of exudates" / "leakage of fluid within the central macula from microaneurysms" / "presence of exudates within the radius of one disc diameter from the macula center" |
| no referable diabetic macular edema               | "no apparent exudates"                                                                                                                                                                             |
| hard exudates                                     | "small white or yellowish deposits with sharp margins" / "bright lesion"                                                                                                                           |
| soft exudates                                     | "pale yellow or white areas with ill-defined edges" / "cotton-wool spot" / "small, whitish or grey, cloud-like, linear or serpentine, slightly elevated lesions with fimbriated edges"             |
| microaneurysms                                    | "small red dots"                                                                                                                                                                                   |
| haemorrhages                                      | "dense, dark red, sharply outlined lesion"                                                                                                                                                         |
| non clinically significant diabetic macular edema | "presence of exudates outside the radius of one disc diameter from the macula center" / "presence of exudates"                                                                                     |
| age-related macular degeneration                  | "many small drusen" / "few medium-sized drusen" / "large drusen"                                                                                                                                   |

# Sigmoid loss: Accommodating multiple labels



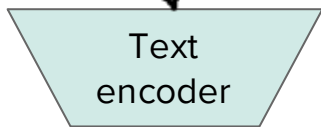
Li et al. Learning transferable visual models from natural language boosting medical vision understanding from multi-granular language learning, **ICLR'26**

Zhai et al. Sigmoid loss for language image pre-training, **ICCV'23**

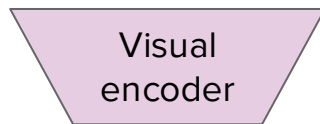
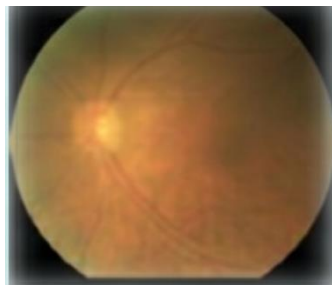
# Embedding prior knowledge about concept hierarchies/relationships

## Level 1: image biomarkers

“few microaneurysms,  
haemorrhage, few hard  
exudates”



$\mathbf{z}_j$

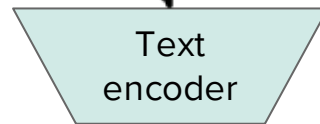


$\mathbf{v}_i$

$$-\log \frac{1}{1 + \exp^{x_{ij}(-t\mathbf{v}_i^\top \mathbf{z}_j + b)}}$$

## Level 2: Disease diagnosis

“Mild diabetic retinopathy”



$\mathbf{z}_l$

$$-\log \frac{1}{1 + \exp^{x_{il}(-t\mathbf{v}_i^\top \mathbf{z}_l + b)}}$$

**Predictions should be consistent for related concepts**

# Embedding prior knowledge about concept hierarchies/relationships

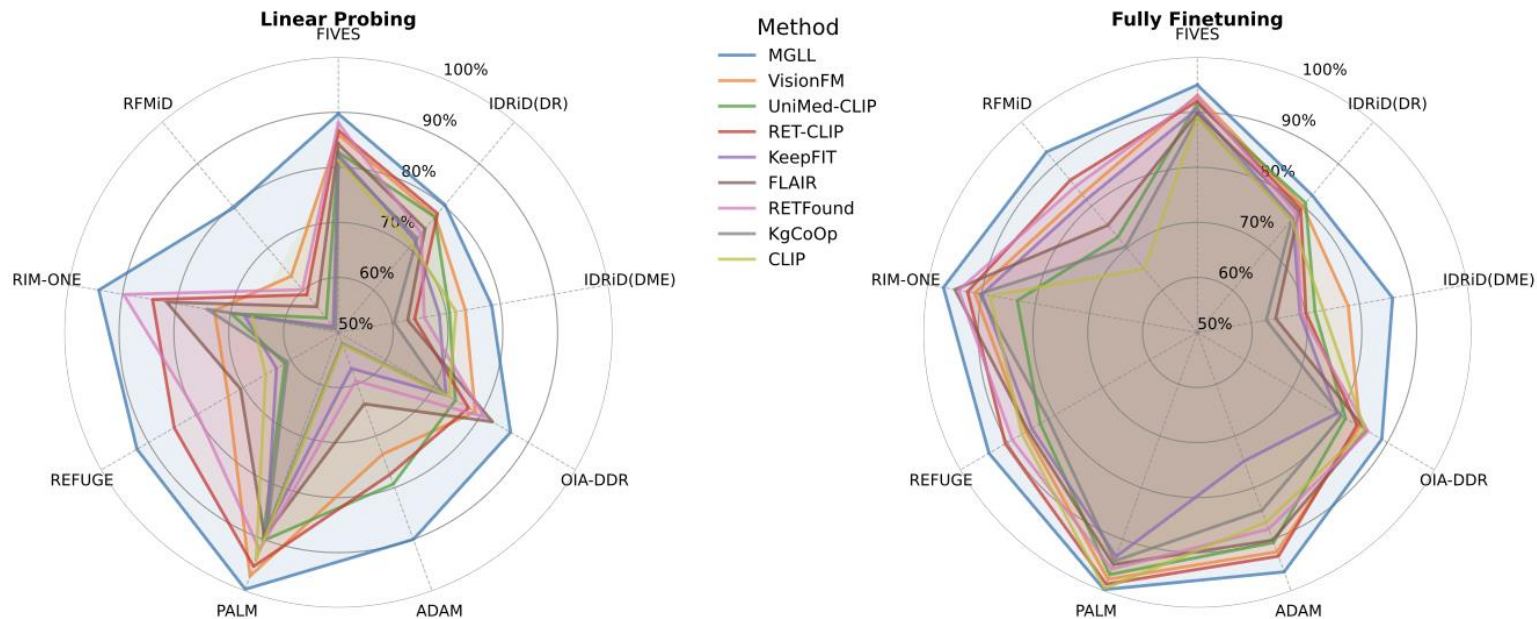


Figure and results from [Li et al. Learning transferable visual models from natural language boosting medical vision understanding from multi-granular language learning, **ICLR'26**]

# Embedding prior knowledge about concept hierarchies/relationships

| Method                          | AMD           | Cataract      | CSR           | DR            | Glaucoma      | Media Haze    | Myopia        | Retinitis     | DME           | Tessellation  | Average ↑                |
|---------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------------------|
| InstructBLIP Dai et al. (2023)  | 80.17%        | 80.00%        | 0.00%         | 76.51%        | 59.30%        | 16.13%        | 44.25%        | 11.11%        | 63.79%        | 41.67%        | 47.29%                   |
| + MGLL                          | <b>83.63%</b> | <b>85.00%</b> | <b>28.57%</b> | <b>82.55%</b> | <b>65.43%</b> | <b>45.16%</b> | <b>52.65%</b> | <b>44.44%</b> | <b>74.14%</b> | <b>58.33%</b> | <b>61.99%</b> (14.7% ↑)  |
| Mini-Gemini Li et al. (2024)    | 76.61%        | 85.00%        | 14.29%        | 79.87%        | 67.90%        | 38.71%        | 58.41%        | 33.33%        | 60.34%        | 33.33%        | 54.78%                   |
| + MGLL                          | <b>82.46%</b> | <b>85.00%</b> | <b>42.86%</b> | <b>84.56%</b> | <b>72.22%</b> | <b>58.06%</b> | <b>64.16%</b> | <b>55.56%</b> | <b>65.52%</b> | <b>41.67%</b> | <b>65.21%</b> (10.4% ↑)  |
| Qwen-VL Bai et al. (2023)       | 81.87%        | 75.00%        | 28.57%        | 80.54%        | 78.40%        | 54.84%        | 76.55%        | 22.22%        | 84.48%        | 25.00%        | 60.75%                   |
| + MGLL                          | <b>85.96%</b> | <b>80.00%</b> | <b>42.86%</b> | <b>89.93%</b> | <b>87.04%</b> | <b>70.97%</b> | <b>80.97%</b> | <b>33.33%</b> | <b>89.66%</b> | <b>41.67%</b> | <b>70.24%</b> (9.5% ↑)   |
| InternVL Chen et al. (2024a)    | 81.29%        | 85.00%        | 71.43%        | 94.63%        | 89.51%        | 64.52%        | 88.05%        | 44.44%        | 87.93%        | 66.67%        | 77.35%                   |
| + MGLL                          | <b>86.55%</b> | <b>90.00%</b> | <b>71.43%</b> | <b>96.64%</b> | <b>90.74%</b> | <b>67.74%</b> | <b>91.15%</b> | <b>55.56%</b> | <b>94.83%</b> | <b>75.00%</b> | <b>81.96%</b> (4.6% ↑)   |
| LLaVA Liu et al. (2023)         | 83.04%        | 90.00%        | 42.86%        | 87.25%        | 91.36%        | 48.39%        | 88.50%        | 44.44%        | 93.10%        | 58.33%        | 72.73%                   |
| + MGLL                          | <b>84.80%</b> | <b>90.00%</b> | <b>57.14%</b> | <b>93.96%</b> | <b>91.98%</b> | <b>61.29%</b> | <b>90.71%</b> | <b>66.67%</b> | <b>96.55%</b> | <b>66.67%</b> | <b>79.98%</b> (7.3% ↑)   |
| LLaVA-Med Li et al. (2023)      | 16.37%        | 15.00%        | 42.86%        | 26.85%        | 25.31%        | 25.81%        | 23.89%        | 33.33%        | 16.67%        | 16.67%        | 24.28%                   |
| + MGLL                          | <b>58.48%</b> | <b>65.00%</b> | <b>57.14%</b> | <b>77.18%</b> | <b>59.26%</b> | <b>51.61%</b> | <b>57.08%</b> | <b>44.44%</b> | <b>55.17%</b> | <b>58.33%</b> | <b>58.37%</b> (34.1% ↑)  |
| Med-Flamingo Moor et al. (2023) | 25.73%        | 30.00%        | 57.14%        | 36.91%        | 24.07%        | 22.58%        | 18.58%        | 22.22%        | 24.14%        | 8.33%         | 26.97%                   |
| + MGLL                          | <b>69.01%</b> | <b>75.00%</b> | <b>71.43%</b> | <b>80.54%</b> | <b>61.11%</b> | <b>54.84%</b> | <b>45.58%</b> | <b>44.44%</b> | <b>51.72%</b> | <b>33.33%</b> | <b>58.70%</b> (31.7% ↑)  |
| Janus-Pro Chen et al. (2025)    | 88.30%        | 75.00%        | 42.86%        | 93.29%        | 90.74%        | 58.06%        | 87.17%        | 33.33%        | 62.07%        | 58.33%        | 68.92%                   |
| + MGLL                          | <b>90.64%</b> | <b>85.00%</b> | <b>71.43%</b> | <b>96.64%</b> | <b>95.06%</b> | <b>67.74%</b> | <b>90.27%</b> | <b>55.56%</b> | <b>70.69%</b> | <b>75.00%</b> | <b>79.80%</b> (10.88% ↑) |

Table and results from Li et al. Learning transferable visual models from natural language boosting medical vision understanding from multi-granular language learning, **ICLR'26**

# We are currently building a million-scale image-text data across hundreds of clinics in Canada

